Supplementary Information (Online Resource 1)
for
Strong Tree-Childness Is a Sharp Generic-Identifiability Boundary for Level-2 Jukes–Cantor Networks

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1 Purpose and scope

This supplement is a reader-oriented map from the mathematical statements in the article to the exact reproducibility package. It does not broaden the theorem. The positive result concerns binary, LSA-valid, already-simple, reticulation-preserving semi-directed strongly tree-child level-2 networks under the open four-state Jukes–Cantor model

$$0 < x_e < 1, \qquad 0 < \lambda_r < 1.$$

Admissible rootings reduce to one fixed mixed graph after the single root suppression in Definition 2.1 of the article; arbitrary hidden rooted refinements are not included.

The relation $N \preceq_{\mathrm{JC}} N'$ means that a source-regular common point has a relatively open neighborhood in the regular source-model locus contained in the target image. The relation $N \bowtie_{\mathrm{JC}} N'$ requires a full-dimensional regular stochastic germ in both directions. Neither relation means equality of complex model closures or equality of the complete open stochastic images.

2 Theorem and proof map

The main theorem states

$$N \preceq_{\mathrm{JC}} N' \quad \Longleftrightarrow \quad \mathcal{C}(N, N'),$$

where $\mathcal{C}(N, N')$ means that the labelled reduced bridge trees agree and corresponding nontrivial blobs are labelled-isomorphic or differ by ordinary triangle redirection T. The same condition characterizes $N \bowtie_{\mathrm{JC}} N'$, so there is no proper one-sided generic containment. The logical dependencies are:

P1. The fixed-mixed-graph convention and no-omnian criterion identify the topology class and imply that a level-2 strong blob has at most one triangle.

P2. Exact Fourier flattening minors recover every cut split pointwise, including the crossing case with two active endpoint tensors. A run-order lemma reduces every four-through-eight-port binary word to a direct obstruction or a certified finite palette.

P3. Positive rank-one cut factorization recovers the bridge tree's local tensors only modulo the full incidence-scaling action. It does not recover physical bridge multipliers. The physical fibre is the ambient incidence orbit intersected with the complete open realization locus, including every local-arm, bridge, and inheritance constraint. Every component tensor is normalized by $P_v(0, \ldots, 0) = 1$; this excludes an additional constant scalar orbit that is not part of the incidence action. Analytic incidence slices give a local product chart in projective local tensors and normalized effective bridge scales.

P4. Every nontrivial level-2 factor contracts to a cycle or one of four directed theta cores.

P5. A decorated graph-to-polynomial atlas classifies every rigid bounded relation in both containment directions. The only nonisomorphic common local germ is ordinary triangle redirection.

P6. A semialgebraic marginal open-image lemma and coherent one- and two-port probes restore every arbitrary subdivision word from the bounded support.

P7. Projective localization turns global containment into independent local containments; the bridge tree has no scaling holonomy.

P8. Conversely, physical analytic sections and uniformly small effective scales glue finitely many local triangle germs simultaneously through the common bridge tree, proving the global common germ.

For a fixed source topology, the singular locus, lower-rank images, wrong topology intersections, and certified nonzero witness hypersurfaces form a proper algebraic exceptional subset. Outside it, the exact distribution determines the labelled topology modulo T.

3 Exact convention checks

The separately implemented convention checker reconstructs rooted bidegrees, acyclicity, the lowest-stable-ancestor condition, one-step root suppression, mixed-graph simplicity, retained arrowheads, admissible rootings, and the tree-child property. Its relevant exact checks are:

Object	exact count or outcome
Primitive support mixed graphs	12
Admissible rootings of those supports	100
Tree-child rootings of the simple double-triangle $K_4 - e$ core	0
Hidden one-step zipper admitted as a rooting of the final tree	no
Nested hidden zipper admitted as a rooting of the final tree	no

The last two regressions prevent the rejected cleanup-fibre convention from re-entering the active theorem. Exhaustive cleanup remains permitted only in explicit restriction or displayed-network operations; it does not enlarge the admissible-rooting set of a fixed mixed graph.

4 Local classification certificates

The finite theorem object is a directed decorated relation, not a target graph identifier. Each normalized record retains both mixed graphs, side colours, independently distinguished incoming boundaries, physical port matching, and source-to-target direction. Across the branch-specific records, the verifier reconstructs and validates every vertex, edge, inheritance, label, and Fourier-coordinate transport needed by that branch; the serialized records retain and content-bind the transport data actually consumed downstream. The verifier regenerates the chain

$$\text{graphs} \to \text{switchings} \to \text{descendant masks} \to \text{Fourier tensors} \to \text{pullback} \to \text{sign/rank certificate}.$$

The following counts are checksums of the regenerated universe rather than premises of the proof.

Certificate family	exact count
Raw three-outgoing decorated presentations	10,826
Canonical three-outgoing directed relations	10,466
Strict three-outgoing relations	5,284
Distinct three-outgoing strict polynomial bodies	800
Pending three-outgoing canonical relations	5,120
Restoration-root raw coverages	5,344
Three-outgoing restoration states	68,584
Generic / refined / strict restoration states	56,055 / 8,349 / 4,036
Three-outgoing isomorphism / T terminals	120 / 24
Direct residual anchors	62
Four-outgoing completion records	6,138
Four-outgoing raw survivors	192
Direct / rooting-duplicate / restoration-root survivors	18 / 42 / 132
Four-outgoing direct quotient classes	3
Four-outgoing nonretaining quotient classes	57
Four-outgoing restoration states	2,106
Four-outgoing generic / refined / isomorphism states	1,860 / 114 / 132
One-port direct-anchor probes	2,642
Two-port direct-anchor probes	18,224
Three-outgoing compact path-bound probes	101,148
Four-outgoing compact path-bound probes	168,582

The evidence map has one row per three-outgoing canonical directed relation and one row per four-outgoing normalized survivor presentation. In the four-outgoing rows the presentation identifier and the further quotient- canonical digest are separate fields, so all 192 presentation identities and their multiplicities remain visible. The serialized transport streams retain the 132 restoration-root transports and 42 duplicate-presentation transports; the 18 direct survivors are instead bound by their direct classification records. These records form 3 direct and 57 nonretaining quotient classes.

Every strict relation has a regenerated exact polynomial, flattening minor, or Bernstein-basis sign certificate on the complete open cube. Every equality terminal is independently canonicalized and accepted only as a labelled isomorphism or ordinary T. Mutation tests reject deleted relations, merged source embeddings, altered port matches, source–target reversal, separator reassignment, broken transports, missing hard-cover keys, and incoherent probes.

For cut-word exhaustiveness, the standalone reduction verifier enumerates all 808,642 balanced binary word distributions with four through eight active ports. It assigns 229,988 to the direct three-run path obstruction and 578,654 to the direct or singleton-doubled short palette, with no unclassified record. Independent graph/switching implementations agree on all 379,742 valid reduced-palette presentations and find no survivor.

For every nonordinary three-port endpoint, the exact code identifies the unique complete central singleton-signature edge class and sets its effective product to one before evaluating $\Delta = abc - t^2$ and $\Gamma = a - bc$. The normalized census has 67 cases with $\Delta > 0$, 2 with $\Delta = 0, \Gamma > 0$, and 7 with $\Delta = \Gamma = 0$; the ordinary trivalent endpoint adds one further zero-zero case. Thus the certificate proves precisely the weak dichotomy used in the article. The JC state group is denoted by $\mathcal{G} = \mathbb{Z}_2^2$.

The marginal-submersion replay groups complete zero-sum split/complement-normalized switching rows, discards tensor-invisible rows, constructs the product-map Jacobian over $\mathbb{Q}$, and verifies a nonzero minor for all 42,908 bounded completions. The auxiliary probe census honestly records 372 one-port ambiguity groups: these are the reason that two-port probes are required, not failures of coherence. The load-bearing direct-anchor and compact semantic gates verify every exact two-port parent transport and reject altered pair orders.

5 Triangle-free Omega sharpness certificate

The historical four-leaf Omega pair was admitted only after a bounded primary and clean-room audit. Each fixed mixed graph is simple, binary, triangle-free, level two, and has seven admissible rootings, exactly two of which are tree-child. Hence both lie in $W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$. The source and target are labelled nonisomorphic: leaf 1 has unordered distances $\{3, 3\}$ to the reticulations in the source and $\{2, 4\}$ in the target.

The exact nine-variable rational correspondence annihilates all 64 zero-sum Fourier differences. At its certified point, all 256 Fourier entries and all 256 inverse-pattern probabilities agree and every parameter is strict. All four rank-nine witnesses use orbit rows $(A, B, C, D, E, F, G, H, K)$. In the declared parameter order $(e_0, \dots, e_{11}, \lambda_V, \lambda_X)$, their columns and exact determinants are as follows; no uniqueness of these minors is claimed. The displayed-rooting columns use the ordered arc list in the article's exact Omega correspondence. For manual reproduction, the alternative rooting (historical census entry N26) uses the ordered arc array

$$\begin{gathered} U \to P0\_0,\ P0\_0 \to P0\_1,\ P0\_1 \to V,\ S \to U,\ S \to V,\ U \to X0, \\ V \to P4\_0,\ P4\_0 \to X0,\ P0\_0 \to L0,\ P0\_1 \to L1, \\ P4\_0 \to L2,\ X0 \to L3. \end{gathered}$$

The source assigns labels $(1, 2, 3, 4)$ to $(L0, L1, L2, L3)$ and the target assigns $(2, 1, 4, 3)$. In the declared order $(e_0, \dots, e_{11}, \lambda_V, \lambda_{X0})$, the strict common parameter vectors for this alternative rooting are

$$\begin{aligned} N26_{\text{source}} :& \quad \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{2}{3}, \frac{1}{4}, \frac{1}{2}, \frac{1}{20}, \frac{1}{2}, \frac{1}{2}, \frac{1}{10}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right), \\ N26_{\text{target}} :& \quad \left(\frac{1}{7}, \frac{1}{2}, \frac{41}{48}, \frac{19}{24}, \frac{14}{19}, \frac{14}{41}, \frac{1}{2}, \frac{12}{205}, \frac{1}{2}, \frac{1}{2}, \frac{3}{40}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right). \end{aligned}$$

The arc arrays and these parameter vectors are stored in `sharpness/omega/inputs/jc_omega_move.json`; `sharpness/omega/cleanroom/verify_omega_release.py` regenerates the complete exact replay record.

Model point	Parameter columns (zero based)	Determinant
Displayed-rooting source	$(0, 1, 2, 3, 4, 7, 8, 9, 10)$	$-171/2305843009213693952000000$
Displayed-rooting target	$(0, 1, 2, 3, 4, 7, 8, 9, 10)$	$-513/9223372036854775808000000$
Alternative-rooting source	$(0, 1, 2, 3, 5, 7, 8, 9, 10)$	$57/576460752303423488000000$
Alternative-rooting target	$(0, 1, 2, 3, 5, 7, 8, 9, 10)$	$189/2305843009213693952000000$

The displayed-rooting source and target minors give the rank lower bound and regularity used in the proof. The alternative-rooting minors are an additional exact check that this certificate is invariant under the admissible rooted presentation. With the four pendant multipliers set to one, the core Jacobian has shape 14×10 and exact rational-function rank six. In the edge order of the paper, at the strict source core coordinates in the pendant-one gauge, the orbit rows (A, B, C, D, E, F) and columns $(a_0, a_1, a_2, a_3, a_4, a_7)$ have determinant

$$-\frac{723}{8589934592} \neq 0$$

at the strict source point. Four pendant torus directions add at most four dimensions, while

$$a_4 \partial_{a_4} c_g + a_7 \partial_{a_7} c_g = \mathbf{1}[g_4 \neq 0] c_g$$

identifies the leaf-4 pendant direction with a core tangent direction. This gives the exact upper bound nine. Thus both complete models and their local intersection have dimension nine. Repeated identical cherry substitution has a positive analytic inverse, adds two dimensions per leaf, and gives the triangle-free family of dimension $2n + 1$ for every $n \geq 4$.

The clean-room release verifier rejects all twelve mandated mutations, including incomplete rooting enumeration, a changed arrowhead or parent, boundary parameters, a changed regenerated Fourier coordinate, and a replaced rank-nine minor.

6 Triangle-containing Theta sharpness certificate

The separately implemented frozen Theta package verifies a second mechanism. Its two four-leaf theta graphs are nonisomorphic and not T-equivalent, lie in $W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$, agree on all 256 Fourier coordinates at a strict quadratic point, and have nonzero rank-eight minors. The common irreducible JC model has dimension eight. The same cherry inverse extends this family to every $n \geq 4$, with common dimension $2n$. Theta is not used as a move in the positive strong-class proof.

For completeness, here are both physical points and the Jacobian gauges. For the source K_4, the edge multipliers are

arc	multiplier	arc	multiplier
$\rho \to A$	$2/3$	$\rho \to C$	$3/4$
$A \to B$	$3/5$	$B \to C$	$1/2$
$C \to D$	$9/20$	$D \to E$	$2/5$
$A \to F$	$1/2$	$E \to F$	$1/3$
$B \to 1$	$1/5$	$D \to 2$	$1/2$
$F \to 3$	$1/2$	$E \to 4$	$3/8$

Both source inheritance probabilities are $1/2$.

Let β be the smaller root of

$$43337075\beta^2 - 36083110\beta + 7336259 = 0, \qquad \frac{441}{1250} < \beta < \frac{3529}{10000}.$$

In the arc notation of Section 9 of the article, the target multipliers are

arc	multiplier	arc	multiplier
$\rho \to A$	$2/3$	$\rho \to C$	$3/4$
$A \to B$	$\dfrac{24835\beta}{20678 - 24835\beta}$	$B \to C$	$1/2$
$C \to D$	$9934/12215$	$D \to E$	$171/775$
$A \to F$	$\dfrac{10339}{53010\beta}$	$E \to F$	$1/2$
$E \to 1$	$31/190$	$D \to 2$	$1/2$
$F \to 3$	$1767/4832$	$B \to 4$	$\dfrac{3}{20\beta}$

Both inheritance probabilities are $1/2$. The root-edge factorization $(2/3)(3/4) = 1/2$ realizes the effective multiplier used after root suppression. The isolating interval gives

$$0 < \beta < \frac{20678}{2 \cdot 24835}, \qquad \beta > \max\left\{\frac{10339}{53010}, \frac{3}{20}\right\}.$$

These inequalities put the three nonrational entries strictly in $(0, 1)$; all remaining entries are visibly strict. Substitution into the two factorized rank-eight determinants printed in the article gives nonzero values, because every listed factor is nonzero on this interval. The frozen quadratic-field verifier separately reduces all 256 Fourier-coordinate differences to zero.

For the source determinant, fix

$$x_{AC} = x_{AF} = x_{BC} = \lambda_C = \lambda_F = \frac{1}{2}$$

after root suppression and use the ordered free variables

$$(P, s, Q, t, R, u, v, S) = (x_{D2}, x_{DE}, x_{E4}, x_{EF}, x_{F3}, x_{CD}, x_{AB}, x_{B1}).$$

For the target determinant, fix

$$x_{AC} = x_{EF} = x_{BC} = \lambda_C = \lambda_F = \frac{1}{2}$$

and use

$$(P', x, y, z, R', w, S', Q') = (x_{D2}, x_{DE}, x_{AF}, x_{AB}, x_{F3}, x_{CD}, x_{E1}, x_{B4}).$$

With these definitions, the two factorizations printed in the article are literal Jacobian determinants of $(A, \ldots, H)$ in the displayed variable orders. The source values are

$$(P, s, Q, t, R, u, v, S) = \left(\frac{1}{2}, \frac{2}{5}, \frac{3}{8}, \frac{1}{3}, \frac{1}{2}, \frac{9}{20}, \frac{3}{5}, \frac{1}{5}\right),$$

and the target values are the corresponding entries of the target table.

7 Deterministic reproduction

Extract the curated certificate archive, enter its top-level directory, and run

```
bash verify.sh quick
bash verify.sh full
bash verify.sh regenerate-all
```

The archive filename is `stc_jc_sharp_boundary_atlas_certificates_v1.1.7.tar.gz`; its Zenodo DOI is `ZENODO_DOI_PENDING` until the depositor reserves it. This token must be replaced before submission. The archive's external SHA-256 authenticates the complete compressed object. After extraction, the quick command authenticates every proof-payload file, checks that `SHA256SUMS` is the exact projection of `ACTIVE_MANIFEST.json`, and checks the frozen theorem summaries. The full command replays every load-bearing exact component certificate. The regeneration command performs two isolated rebuilds from primitive graph encodings and requires their logical commitments to agree exactly.

The whole-proof checker is a fail-closed consistency gate, not a substitute for the component calculations. The mathematical evidence is supplied by the graph-derived component scripts and their separately implemented replays. The supplied transcripts record the complete commands and outputs; the pinned runtime versions are stated in `environment/exact_tool_versions.txt` and `RUNTIME_AND_HARDWARE.md`.

8 Theorem-to-artifact crosswalk

Claim	Certificate-bundle-relative evidence
Convention, primitive supports, cut recovery, and bridge fibre	`primary/certificates/core_universe.json`, `primary/certificates/support_universe.json`, `independent/bridge_cut/palette_reduction_certificate.json`, `reviews/global_bridge/palette_cleanroom_certificate.json`, the remaining `independent/bridge_cut/` certificates, and `reviews/global_bridge/exact_audit_certificate.json`.
The complete three- and four-outgoing finite atlas, restoration, and probes	`atlas/ATLAS_EVIDENCE_BINDINGS.jsonl.gz`, whose theorem rows point to `atlas/RESTORATION_CLOSURE_BINDINGS.jsonl.gz` or `atlas/DIRECT_ANCHOR_CLOSURE_BINDINGS.jsonl.gz`; restoration terminals point onward to `atlas/COMPACT_PATH_CLOSURE_BINDINGS.jsonl.gz`, binding every exact path, transport, witness, and polynomial record.
Complete deterministic regeneration from primitive inputs	`verifiers/regenerate_load_bearing.py`, invoked twice by `verify.sh` in `regenerate-all` mode.
Ordinary triangle common germ	`primary/certificates/jc_triangle_redirection_active.json` and `reviews/triangle_redirection_cleanroom/`.
Omega and Theta sharpness families	`sharpness/omega/` and `sharpness/theta/`.
Integrity, totality, deterministic regeneration, and mutations	`ACTIVE_MANIFEST.json`, all four `atlas/*BINDINGS.jsonl.gz` streams, `expected_outputs/`, and `verify.sh`.

9 Interpretation limits

The theorem reconstructs topology, not all numerical parameters. In particular, bridge data determine projective local tensor orbits and normalized effective scales, not physical bridge lengths.

The ordinary triangle result is a common full-dimensional regular germ; it does not assert equality of complete stochastic images or one orientation compatible with every distribution. No pointwise classification of the exceptional locus, richer substitution model, finite-sample inference result, or complete weak-class classification is claimed.